Purwar, who joined IIT Bombay as a professor that year, began to use the institute's lab to set up an immunoengineering team that would spend the next 3-4 years developing CD-19 CAR-T, the country's first indigenous chimeric antigen receptor T-cell therapy—a treatment where human T cells (a kind of white blood or immune cell) are altered in the lab to hunt and attack cancerous cells. Today, CD-19 CAR-T has received two patents, approval in the US to treat certain blood cancers and is gearing up for its Phase 2/ 3 trials in India, and there is hope that it could be offered as a treatment in Indian hospitals by early next year. "It is an exciting time for gene and cell therapy in India," says Purwar, who is now the CEO of ImmunoACT, the company that will make the new therapy available in India once it is approved at a fraction of the cost overseas. "There is a lot more investment and interest and the results are very encouraging. We are estimating cost to be around Rs 30 lakh per patient compared to Rs 3-4 crore in the US," he says. This should offer a new lease of life to people who earlier thought cancer was the end of the road for them.

Cancer is a disease where abnormal cells divide uncontrollably, and in doing so, destroy healthy tissue, eventually claiming a person's life. All cancers start inside cells due to genetic changes within them. Usually, cells produce signals to control how much and how often they divide. If any of these signals turns faulty, cells start to grow without control, eventually resulting in a lump or tumour. In cancer, the body is unable to recognise and dispel the cancerous cells in the same way that it can detect invading viruses or bacteria.



"So far, there were only three options for cancer: surgery, chemotherapy and radiation. If these did not work, you had no choice: cancer was your death sentence"

RAHUL PURWAR
CEO, ImmunoACT

WHAT ARE CANCERS AND WHY DO THEY OCCUR?

- All cancers start inside cells due to genetic changes within them. Usually, cells produce signals to control how much and how often they divide. If any of these signals turn faulty, cells start to grow without restraint, eventually forming a lump or tumour. The body is unable to detect and dispel the cancerous cells in the same way that it can ward off invading viruses or bacteria
- For some cancers, the cause is already known. Cervical cancer, for example, is caused by a human papillomavirus infection. For most other cancers, the exact cause of the genetic changes within cells remains unknown

Graphics by TANMOY CHAKRABORTY

Cancer treatment thus far has relied either on radiation or chemotherapy to kill the cancerous cells or surgery to remove the tumour altogether. However, these interventions have had a success rate of less than 10 per cent in advanced stages of cancers such as of the blood. The side-effects of chemo and radiation—fatigue, nausea, body pain, skin and hair changes, insomnia—are another debilitating factor, as both treatments end up harming healthy normal cells alongside the cancerous ones in the body. The current therapies also do not guarantee that every cancer cell is killed off—for advanced-stage cancers, the chance of recurrence remains as high as 30-40 per cent.

An advancing biotechnology industry is dispelling some of this gloom. It is fuelling a new generation of solutions to diagnose, prevent and, in some cases, even cure cancer. From therapeutic antibody drugs that can recognise surface proteins on cancer cells and target them (such as Roche's breast cancer drug PHESGO launched in India last month) to CAR-T therapy to molecular blood tests for early detection of cancer, personalised cancer interventions, many of which were earlier either unaffordable, inaccessible, or simply unbelievable, are changing the whole approach to what renowned oncologist-author Siddhartha Mukherjee poetically described as "the emperor of maladies". India has also developed a new vaccine called Cervavac for cervical cancer, launched by the Pune-based Serum Institute on September 2 (see *A Game-changer Called Cervavac*).

"Cancer as a disease has undergone evolution in treatments and therapies. From surgeries, chemotherapy and radiotherapy to antibody immunotherapies that target certain kinds of cancer antigens, we have come a long way," says Kiran Mazumdar-Shaw, chairperson of Biocon, who along with Siddhartha Mukherjee co-founded Immuneel, an Indian immunotherapy company, in 2019. She points out how the survival rates in some cancers have shot up—breast cancer, for example, had a survival rate of 10 per cent a decade ago; it is almost 90 per cent today if diagnosed in time. "The prognosis for some cancers is so good that you could say you might die with that cancer but you won't die of it," she says.

The promise of these therapies cannot be emphasised enough in a country that has the third-highest burden of cancer in the world after the US and China. India also has the world's highest rate of triple negative breast cancer (an aggressive, fast-growing type where the cancer cells do not